Robotics



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Lecture # 1

Introduction to Robotics

Introduction

Robotics History

- 1922 Czech author Karel Capek wrote a story called Rossum's Universal Robots and introduced the word "Rabota"(meaning worker)
- 1954 George Devol developed the first programmable Robot.
- 1955 Denavit and Hartenberg developed the homogenous transformation matrices
- 1962 Unimation was formed, first industrial Robots appeared.
- 1973 Cincinnati Milacron introduced the T3 model robot, which became very popular in industry.
- 1990 Cincinnati Milacron was acquired by ABB

Robotics History



Three Laws of Robotics (Isaac Asimov in I,Robot 1950)

Asimov proposed three “Laws of Robotics”

and later added the “zeroth law”

- **Law 0:** A robot may not injure humanity or through inaction, allow humanity to come to harm
- **Law 1:** A robot may not injure a human being or through inaction, allow a human being to come to harm, unless this would violate a higher order law
- **Law 2:** A robot must obey orders given to it by human beings, except where such orders would conflict with a higher order law
- **Law 3:** A robot must protect its own existence as long as such protection does not conflict with a higher order law

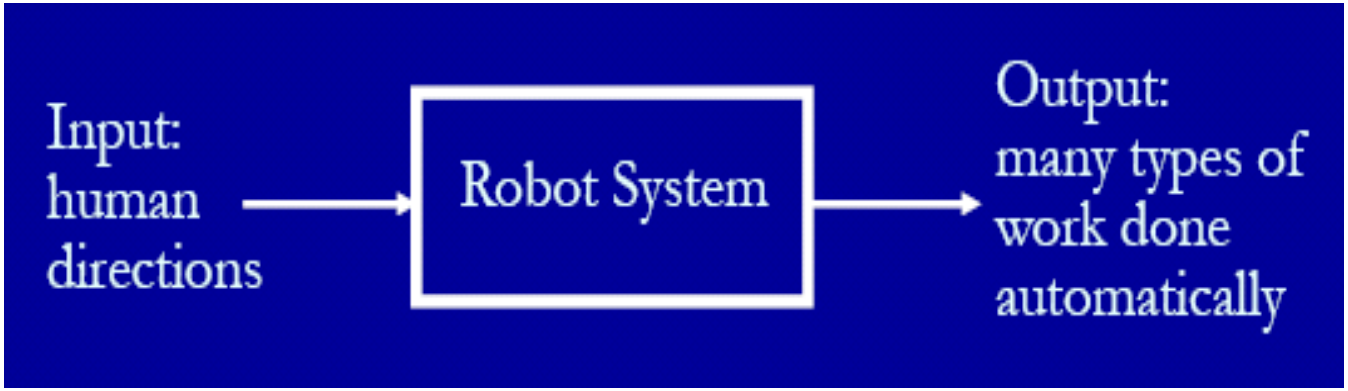
Definition of the Robot

- Defined by **Robotics Industry Association (RIA**) as
  - A Re-programmable, multifunctional manipulator designed to move material, parts, tools or specialized devices through variable programmed motion for a variety of tasks
- **Robotics Association of America**
  - An industrial robot is a re-programmable, multifunctional manipulator designed to move materials, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks.
- **Japan Robot Association (JARA)**
  - Devices capable of moving in a flexible manner analogous to the moving parts of living organisms, with or without intelligent functions, allowing operations in response to human commands

What is a Robot?

In General:

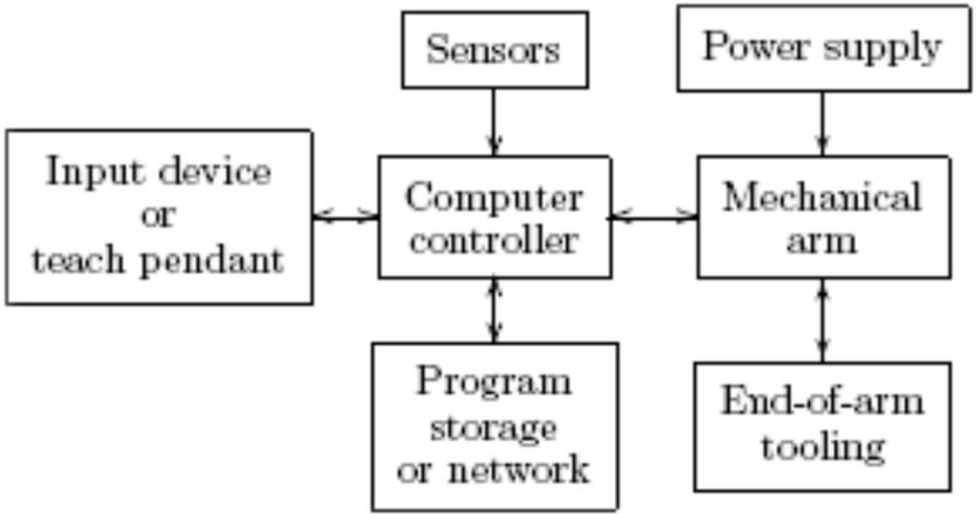
- A machine that senses, thinks, and acts
- Must have sensors, processing ability that emulates some aspects of cognition, and actuators
- In artificial intelligence, such systems are known as “agents.”



Robot System as a Black Box

Robotic System

- Robotics is the art, knowledge base, and the know-how of designing, applying, and using robots in human endeavors.
- Robotics is an interdisciplinary subject that benefits from mechanical engineering, electrical and electronic engineering, computer science, biology, and many other disciplines.



Components of a Robotic System

Robot Classification

Robot Classification

The following is the classification of Robots according to the **JIRA (Japanese Industrial Robot Association)**

- **Class1:** Manual-Handling Device
- **Class2:** Fixed Sequence Robot
- **Class3:** Variable Sequence Robot
- **Class4:** Playback Robot
- **Class5:** Numerical Control Robot
- **Class6:** Intelligent Robot

Robot Classification

The following is the classification of Robots according to the **Robotics Institute of America (RIA)**

- **Variable-Sequence Robot :** A device that performs the successive stages of a task according to a predetermined method easy to modify (Class 3)
- **Playback Robot :** A human operator performs the task manually by leading the Robot (Class 4)
- **Numerical Control Robot :** The operator supplies the movement program rather than teaching it the task manually. (Class 5)
- **Intelligent Robot :** A robot with the means to understand its environment and the ability to successfully complete a task despite changes to the environment. (Class 6)

Robot Classification

AFR (Association FranÇaise de Robotique)

- **Type A:** Manual Handling Devices / telerobotics
- **Type B:** Automatic Handling Devices / predetermined cycles
- **Type C:** Programmable, Servo controlled robot, continuous point-to-point trajectories
- **Type D:** Same type with C, but it can acquire information.

Robot Classification

- **Use**
  - Industrial, Non-industrial
- **Mobility**
  - Fixed, Mobile
- **Power source**
  - Electrically, hydraulically, pneumatically powered
- **Method of control**
  - Non-servo control, Servo-controlled robots
- **Capability**
  - 1st, 2nd, 3rd generation robots
- **Kinematic configuration (geometry)**
  - RRR, RRP, RPP, PPP

Robot Classification

- **First generation**
  - reprogrammable, arm-type, manipulative devices, only capable of memorizing repetitive movements, internal sensors
- **Second generation (1980's)**
  - external sensors, feedback, make limited decisions and respond to the working environment
- **Third generation (late 90's and beyond)**
  - employ artificial intelligence, able to recognize, learn, and think, able to adapt to new situations

Robot Classification

Types of Robots by Function

- Manipulators:
  - robotic arms and hands;
- Motion generators:
  - flight simulators; SCARA (Selective-Compliance Assembly Robot Arm); and moving platforms
- Locomotors, mobile robots:
  - legged and wheeled robots;
- Swimming robots; and
- Flying robots.

Robot Classification

Manipulators / Robotic Arm





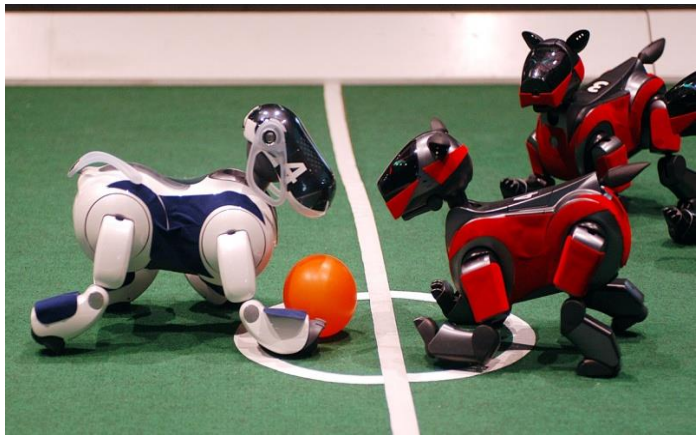






Robot Classification

Legged Robots



Robocup Team



Aibo 4 leged Robot



HEXABOD Robot



Big Dog Robot

Robot Classification

Humanoids



Qurio Humanoid



Asimo Humanoid



NAO Humanoid

Robot Classification

Mobile (Wheeled) Robots

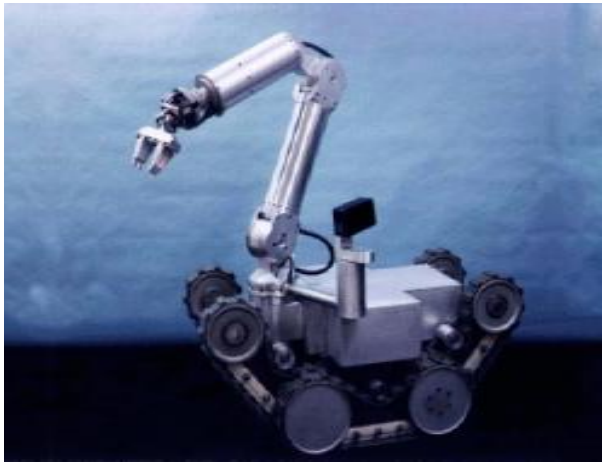




Outdoor Robots



Indoor Robots



HAZBOT



NASA Mars Rover

Robot Classification

Swimming Robots





Autonomous Underwater Vehicle (AUV)
Underwater Robot

Robot Classification

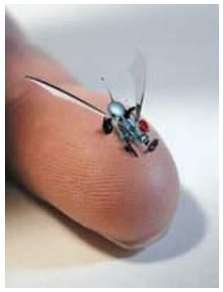
Flying Robots / unmanned aerial vehicles (UAV)



Unmanned Aerial Vehicle



Flying UAV



Micro Robot

Predator



Robot Classification

Robots in Space





NASA Space Station

Robot Classification

Believe or not











Thank you for your attention !

Good Luck !

